COM111 Tutorial 3

1. Solve for the current *i* in the circuit of Fig. 1 using Thevenin's theorem. (*Hint:* Find the Thevenin equivalent seen by the 12 Ω resistor.)

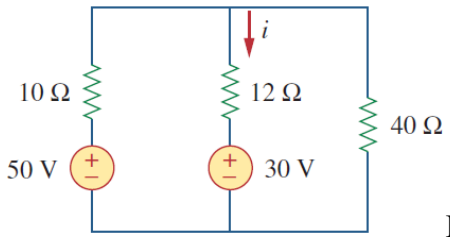


Fig. 1

2. Find the Norton equivalent with respect to terminals *a-b* in the circuit shown in Fig. 2.

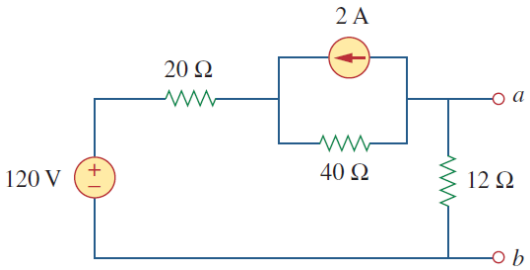


Fig. 2

3. Apply Thevenin's theorem to find in the circuit of Fig. 3.

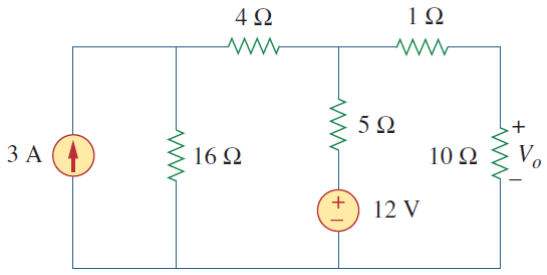


Fig. 3

4. Obtain the Thevenin equivalent at terminals *a-b* of the circuit shown in Fig. 4.

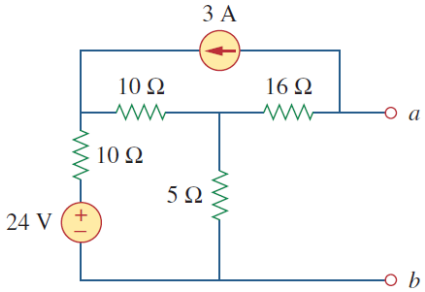


Fig. 4

5. Find the Thevenin equivalent at terminals *a-b* of the circuit in Fig. 5.

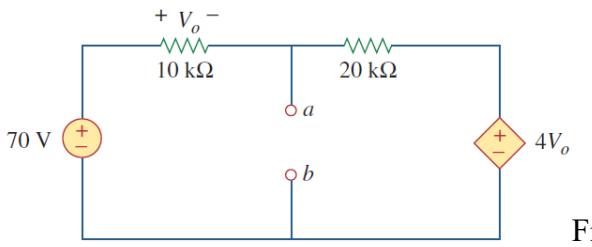


Fig. 5

6. For the circuit in Fig. 6, obtain the Thevenin equivalent as seen from terminals *a-b* and *b-c*:

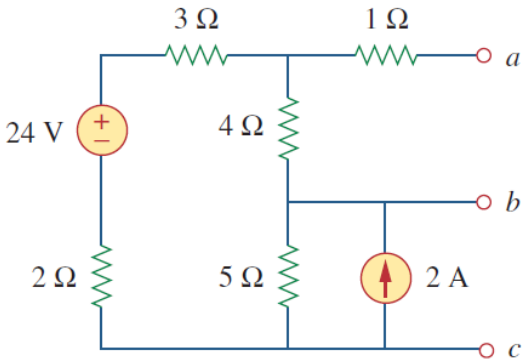


Fig. 6

7. Obtain the Thevenin and Norton equivalent circuits of the circuit in Fig. 7 with respect to terminals *a-b*.

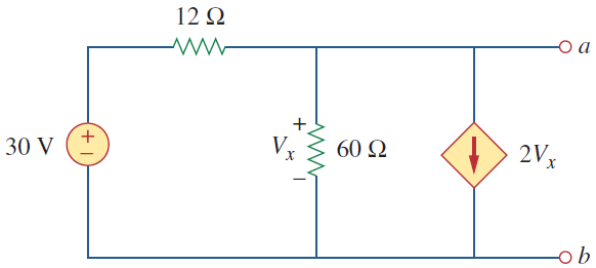


Fig. 7

8. Obtain the Thevenin and Norton equivalent circuits at terminals *a-b* for the circuit in Fig. 8.

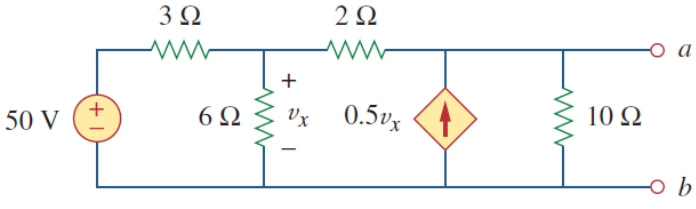


Fig. 8

9. Find the maximum power that can be delivered to the resistor R in the circuit of Fig. 9.

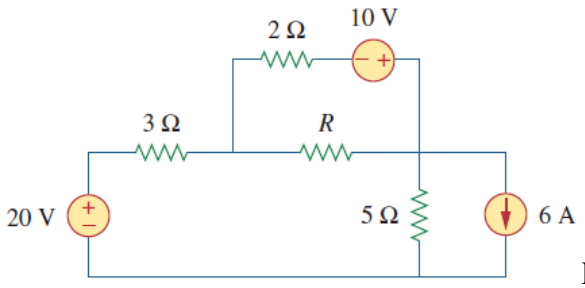


Fig. 9

10. The variable resistor R in Fig. 10 is adjusted until it absorbs the maximum power from the circuit. (a) Calculate the value of R for maximum power. (b) Determine the maximum power absorbed by R.

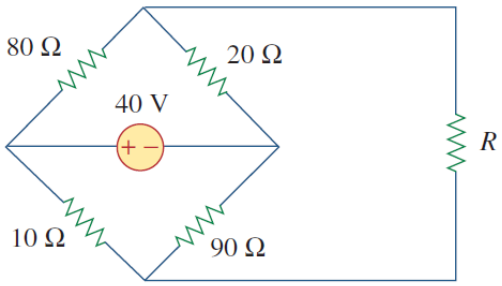


Fig. 10

11. Determine the maximum power delivered to the variable resistor R shown in the circuit of Fig. 11.

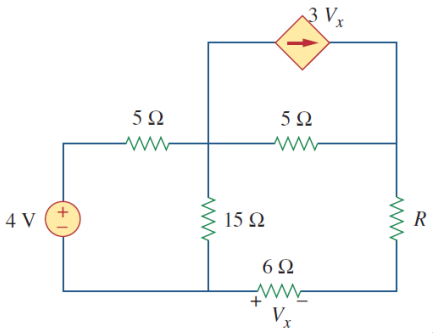


Fig. 11

12. For the circuit in Fig. 12, what resistor connected across terminals *a-b* will absorb maximum power from the circuit? What is that power?

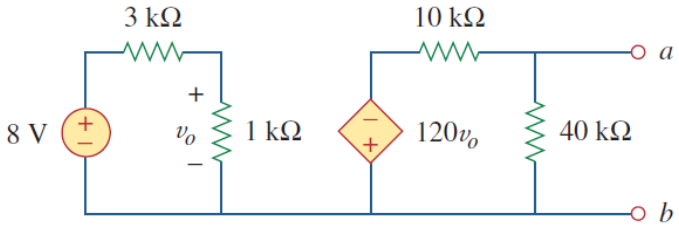


Fig. 12

13. Apply the superposition principle to find in the circuit of Fig. 13.

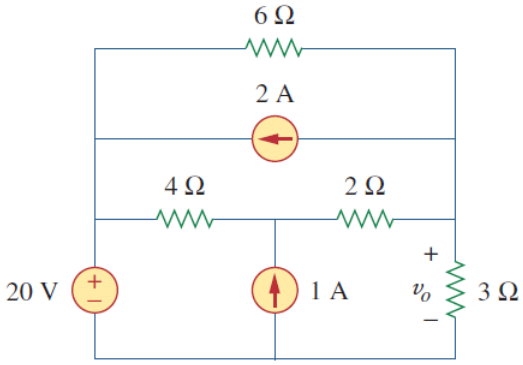


Fig. 13

14. For the circuit in Fig. 14, use superposition to find i. Calculate the power delivered to the 3Ω resistor.

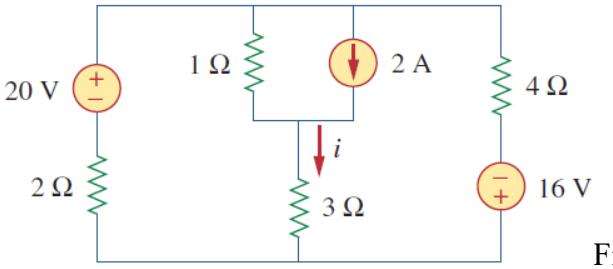


Fig. 14

15. Use superposition to solve for in the circuit of Fig. 15.

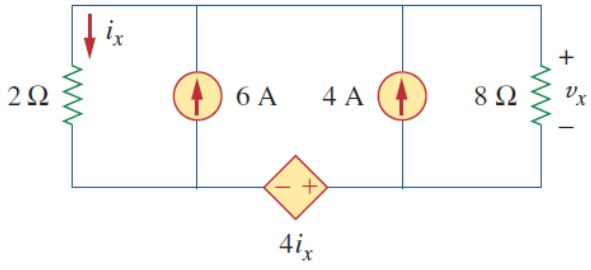


Fig. 15

16. Referring to Fig. 16, use source transformation to determine the current and power absorbed by the 8 Ω resistor.

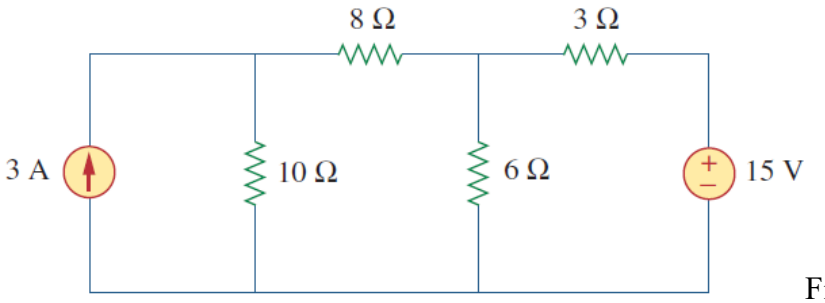


Fig. 16

17. Use source transformation to find the voltage in the circuit of Fig. 17.

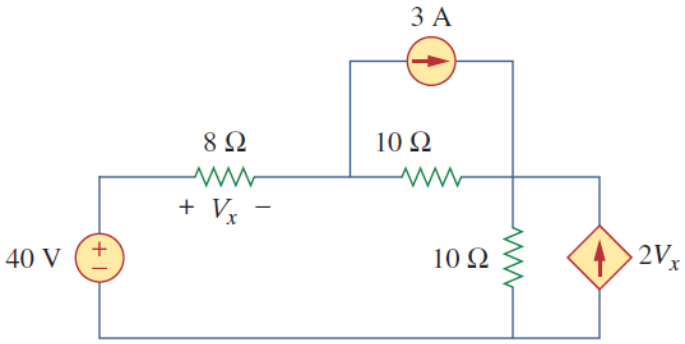


Fig. 17

18. Determine vx in the circuit of Fig. 18 using source transformation.

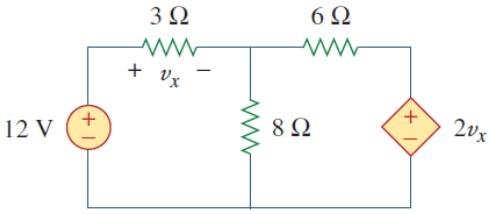


Fig. 18

19. The Wheatstone bridge circuit shown in Fig. 19 is used to measure the resistance of a strain gauge. The adjustable resistor has a linear taper with a maximum value of 100 Ω. If the resistance of the strain gauge is found to be 42.6 Ω, what fraction of the full slider travel is the slider when the bridge is balanced?

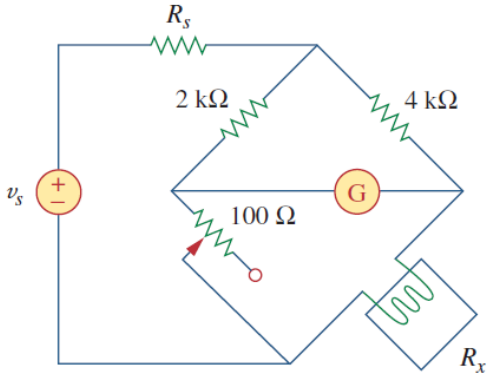


Fig. 19

20. (a) In the Wheatstone bridge circuit of Fig. 20, select the values of R1 and R3 such that the bridge can measure Rx in the range of 0–10 Ω.

(b) Repeat for the range of 0–100 Ω.

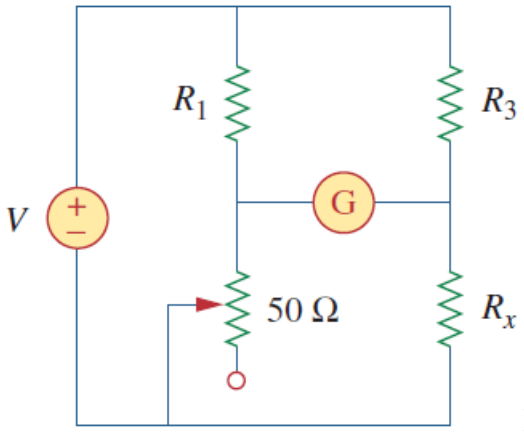


Fig. 20

21. Consider the bridge circuit of Fig. 21. Is the bridge balanced?

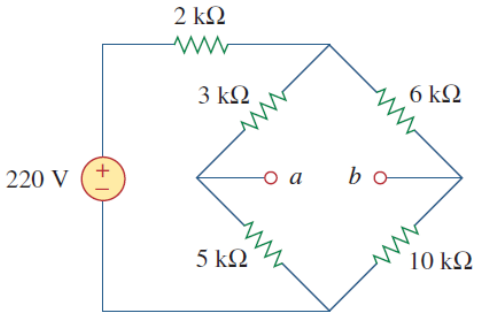


Fig. 21